



Final Submission

How Supply Chain Analytics Can Reshape Fast-Fashion Operations

Supply Chain Analytics - BUAN309

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Case Summary

The Critical Role of Supply Chain Analytics in Fast Fashion

Fast fashion companies work in one of the most complex and time-sensitive supply chain environments in retail. Unlike traditional apparel businesses, fast fashion is built on short product life cycles, a wide variety of SKUs, frequent collection updates, and high seasonality (Business of Fashion, 2023). These factors all increase operational risks significantly.

Even though disruptions from the pandemic have decreased, supply chains in the fashion industry still face significant vulnerabilities. According to Business of Fashion (BoF Insights), leaders in fashion supply chains still deal with ongoing challenges such as rising raw material costs, logistics issues, and unpredictable consumer demand (Business of Fashion, 2023).

A notable real-world example is Dr. Martens, the British footwear brand. The company publicly acknowledged that fixing supply chain problems could cost it up to £11 million in expenses. Additionally, it also spoke of potential losses of £25 million in wholesale revenue because of ongoing supply chain disruptions (Business of Fashion, 2023). This shows how supply chain weaknesses can lead to significant financial losses, even for well-known global fashion brands.

Traditional mass-production models that depend on long planning cycles and bulk inventory are no longer enough. As fashion cycles shorten and consumer preferences change quickly, companies must operate leaner, faster, and more responsively while also managing costs and maintaining service levels.

Supply chain analytics is essential for tackling these challenges. Specifically, analytics enables fashion firms to:

- Identify high-risk SKUs with unpredictable demand or high return rates
- Improve the accuracy of demand forecasts
- Create inventory policies that balance service levels and costs
- Optimize warehouse locations and transportation routes
- Reduce inefficiencies caused by delays

The concept of Quick Response (QR), which was developed by Kurt Salmon Associates in 1986, showed that while apparel production takes only 11 weeks, it used to take up to 66 weeks for products to reach stores due to planning, inventory allocation, and logistics issues. This insight completely changed fast fashion strategies for retailers like Zara and H&M, where speed and responsiveness became key advantages. These studies highlight that analytics is now crucial to not only survive but also thrive in fast fashion.

Project Motivation and Selection

This project focuses on a fast fashion company, XYZ Pvt. Ltd., which is currently growing. As the company expands, it is important to understand supply chain performance to prevent more inefficiencies. The retailer encounters several common challenges in fast fashion, such as highly seasonal demand, specific return issues for different products, widespread delays, and rising logistics and production costs.

While reviewing existing literature and industry case studies, it became evident that there are few publicly available examples that provide a complete view of supply chain analytics for fast fashion, particularly at the factory, warehouse, product, and customer interaction levels. To address this, a synthetic dataset was chosen to keep control over the supply chain structure while ensuring it remains realistic.

The project was chosen for its academic relevance and its close connection to the fashion industry, which is an industry that interests me. The dataset encompasses the entire supply chain process, from manufacturing and warehousing to sales, returns, and delays, allowing for an integrated analytical approach instead of separate functional analysis.

Perspective of the Project

The project is conducted from the viewpoint of BoF Insights, the analytical and consulting division of The Business of Fashion. The aim is not only to assess operational metrics but also to turn analytical findings into strategic recommendations that fashion executives can implement.

The project takes a consulting-style approach to identify performance issues, measure trade-offs, and create innovative solutions using supply chain analytics tools.

The main goal is to create a fully optimized supply chain through demand forecasting, ABC analysis, EOQ, reorder point logic, facility location models (Center of Gravity and p-median), multi-criteria decision analysis, and linear programming-based transportation optimization.

While the importance of supply chain analytics establishes why such an analysis is necessary, the next section describes what data was available to support these decisions.

Data

This project uses a synthetic but realistic dataset that reflects the complexity of a fast-fashion supply chain. The dataset tracks the flow of products from factories to warehouses to customers. This allows for analysis of demand, returns, delays, costs, and network design.

The data covers the period from 1 April 2023 to 25 January 2024, totaling almost 10 months. It includes seasonal changes, such as summer, the festive season, and early winter. This timeframe is important for fast fashion because demand fluctuations and seasonality are key issues.

The dataset is organized into four main tables, with each one representing a different part of the supply chain.

Production Costs

The Production Costs table shows the manufacturing side of the supply chain.

- There are 5 factories, F001 to F005.
- Each factory makes several products, but not every product is made at all factories.

Manufacturing costs vary by:

- Factory
- Product

This shows how fast fashion is made in the real world. Different factories specialize in different product categories. The cost differences arise from labor skills, machinery, scale, and location.

Key variables include:

- Factory_ID
- Product_ID
- Manufacturing_Cost (per unit)

It helps improve transportation by reducing production and shipping costs at the same time.

Warehouse Shipping Costs

The Warehouse Shipping Costs table outlines the logistics between factories and warehouses. There are 20 warehouses, W001 to W020. Shipping costs differ based on:

- Factory
- Warehouse

- Product

Costs are shown per 1,000 pieces, which is typical for bulk apparel shipments.

Key variables include:

- Warehouse_ID
- Factory_ID
- Product_ID
- Shipping_Cost_per_1000_units

Why this is important:

- It helps with transportation cost modeling.
- It allows for comparing long-haul and short-haul routes.
- It supports linear programming for minimizing costs.

This structure reflects real fast-fashion logistics, where:

- Warehouses do not source equally from all factories.
- Shipping costs depend on distance, mode, and product features.

Products

The Products table defines the SKU-level attributes. There are 40 unique SKUs, spanning apparel (jackets, jeans, sweatshirts, sweatpants, blazers) and accessories (gloves, sunglasses, belts, etc.).

Key variables include:

- Product_ID
- Name
- Gender (Men / Women / Unisex)
- Selling_Price
- Weight (kg)

Why this matters analytically:

- Enables gender-based analysis (important for returns).
- Allows testing whether weight or price affects delays or returns.
- Supports revenue and demand calculations.
- Enables ABC analysis based on demand contribution.

This table is critical because fast fashion is SKU-driven:

- Performance varies sharply across products.
- A small number of SKUs often drive the majority of revenue and risk.

Log Data

The Log Data table is the main operational dataset and the most detailed. Each row represents a shipment batch where a quantity of a single product is shipped from one factory to one warehouse on a specific date.

Key variables include:

- Date
- Order_ID
- Product_ID
- Source Factory
- Destination Warehouse
- Shipping Time (Expected)
- Shipping Time (Actual)
- Delay Risk
- Total No. of Pieces
- No. of pieces sold
- No. of Pieces Returned
- Avg. Batch Rating

Derived variables created during analysis:

- Delay (days) = Actual minus Expected
- Is_Delayed (binary)
- Return Rate = Returned divided by Sold
- Monthly period (Year-Month)

This table helps in:

- It enables returns analysis, delay analysis, and sales behavior study.
- It connects operational performance to customer outcomes.
- It allows us to separate supply-side from demand-side issues.

Synthetic Location Design for Warehouses and Factories

Why Synthetic Locations Were Used

The dataset used in this project does not include geographic coordinates (latitude and longitude) for factories or warehouses. However, several supply chain models applied in this study, such as Center of Gravity analysis, p-median facility location modeling, multi-criteria decision analysis (MCDA), and transportation cost optimization, need spatial inputs to work properly.

To use these models without misrepresenting or exposing real company locations, I created synthetic but realistic factory and warehouse locations. The goal was not to copy a specific company's logistics network. Instead, it was to create a realistic illustration of a fast-fashion supply chain working in India.

Warehouse Location Design

Warehouse locations were placed in major Indian consumption and logistics hubs commonly used by large fashion retailers and e-commerce platforms. The selection was based on documented industry practices rather than random placement.

The following criteria were applied:

- High population density and urban consumption demand
- Presence of large fulfillment centers and last-mile logistics infrastructure
- Connectivity to national highways, rail freight corridors, and ports
- Evidence of concentrated apparel and e-commerce warehousing activity

Based on these criteria, warehouses were mapped to the following regions:

- Delhi NCR
- Mumbai/Navi Mumbai (including Bhiwandi)
- Bengaluru
- Hyderabad
- Chennai
- Kolkata
- Ahmedabad
- Pune
- Nagpur
- Jaipur

These locations reflect the actual distribution footprint of major Indian fashion and lifestyle retailers like Myntra, Ajoio, and Reliance Trends, as noted in public industry reports and case studies (Deloitte, 2022; KPMG, 2021; Business of Fashion, 2023).

Each warehouse was assigned approximate coordinates corresponding to these regions. These coordinates do not represent any actual company warehouse but reflect the spatial logic of fast-fashion distribution networks in India.

Factory Location Design

Factory locations were chosen based on established textile and apparel manufacturing clusters in India. The selection mirrors historical and current concentrations of garment production rather than optimized outcomes.

Factories were located in areas known for:

- Availability of skilled apparel labor
- Proximity to raw material and textile processing units
- Long-standing specialization in specific garment categories

Representative factory regions included:

- Tiruppur/Coimbatore – knitwear, T-shirts, sweatshirts
- Bengaluru/Tamil Nadu belt – apparel assembly, jackets
- Noida/Delhi NCR – woven garments, denim
- Surat/Gujarat belt – synthetic fabrics and textile processing
- Ludhiana – winterwear such as gloves, jackets, and hoodies

These clusters are consistently identified in reports published by the Ministry of Textiles (Government of India) and Invest India as major apparel manufacturing hubs (Ministry of Textiles, 2022; Invest India, 2023).

Factories were intentionally placed away from major consumption centers to reflect the typical separation of production and distribution in fast-fashion supply chains.

Distance Calibration and Validation

After assigning factory and warehouse locations, I calculated and reviewed the distances between them to ensure realism. The observed distance ranges were:

- 300–500 km: regional supply movements
- 900–1,200 km: inter-regional distribution
- 1,800–2,200 km: pan-India freight movement

These ranges match recorded freight movement patterns in the Indian apparel sector and are in line with logistics standards found in industry and consulting reports (Deloitte, 2022; Chopra & Meindl, 2016).

The goal of distance in this project was to reduce costs and improve network design, not to forecast performance. This matches earlier research that shows distance has a strong effect on transportation costs, but does not necessarily influence delays or returns in apparel supply chains (Cachon & Swinny, 2011).

Validity and Limitations

The synthetic location design ensures:

- Realistic spatial structure for optimization models
- No use of proprietary or confidential company data
- Meaningful trade-offs between cost, distance, and demand
- Results that are applicable to fast-fashion networks in India

However, certain limitations are acknowledged:

- City-level routing and road congestion are not modeled
- State-level regulations and border delays are not differentiated
- Results are more strategic than operational in nature

For a project of academic nature on a small scale, these limitations are acceptable.

Exploratory Data Analysis

Revenue and Profits

This section looks at how well the fast-fashion supply chain is doing financially. It breaks down revenue, cost, and profit by products, factories, warehouses, time, and customer segments.

Overall Revenue and Profit Performance

Across the analysis period (April 2023, January 2024), the supply chain generated:

- Total Revenue: ₹2,727,609,540
- Total Cost (Production + Shipping): ₹1,116,967,619
- Total Profit: ₹1,610,641,921

This shows that the company is profitable, but costs make up a large portion of revenue, which leaves little room for inefficiency.

Gender

When looking at the data by gender category, men's products show higher profitability:

- Revenue: ₹2.10B
- Profit: ₹1.33B
- Units: ~52.9M

Women's products are important for variety but contribute less:

- Revenue: ₹0.62B
- Profit: ₹0.28B

Unisex products make up a tiny portion of total profit.

This difference shows that men's categories generate most of the economic activity in the supply chain. This is especially true for outerwear and structured garments. While women's categories are important, they may need better cost management, more careful product selection, or different pricing strategies.

SKUs

Profit contribution varies significantly across SKUs, reflecting fast-fashion trends. The top products generating profit are mainly men's outerwear and structured apparel:

- Men's Blazer: ₹210.8M profit
- Men's Jacket: ₹169.2M profit
- Men's Chinos: ₹132.1M profit
- Men's Hoodie: ₹112.7M profit

These items are in high demand and have good cost structures, making them important for profits. However, high revenue doesn't always mean high profit. Men's jeans generate significant revenue but have lower profit margins. This could be due to higher manufacturing costs from using heavier fabric and more complex construction and finishing processes. In contrast, long-tail products like women's jumpsuits, accessories, and niche SKUs barely add to overall profit. This shows why ABC classification is important. Managers should concentrate on a small group of high-impact SKUs instead of dividing their efforts evenly across the entire catalog.

Factory and Warehouse

Profitability varies significantly across factories.

- F001 and F005 are the top contributors, together making up a large portion of total profit.
- F003 and F002 make moderate contributions.
- F004 adds very little, suggesting low volumes, poor cost structures, or limited SKU allocation.

This shows that factory performance is inconsistent. Thus, profit optimization should focus on aligning factories with SKUs, not just reducing costs.

On the other hand, profitability among warehouses is fairly balanced. Most warehouses generate profits within a small range. Warehouses like W011, W015, W004, W005, and W012 contribute the most to profits. Differences in profit are more about volume throughput than major cost issues. Thus, optimizing the network, such as facility selection and routing, is more important than fine-tuning individual warehouses and that the choice of warehouse does not significantly affect returns or delays, but is important for reducing costs and distance.

Monthly Revenue and Profit Trends

Profit trends closely follow seasonal demand. Peak profitability happens in Q4, with December showing the highest profit of ₹178.4M. This increase is driven by winter products like jackets, blazers, and gloves. In contrast, the months in the middle of the year show lower but steady profitability. January sees a noticeable drop, which corresponds with the return to normal demand after the season.

The connection between profit spikes and seasonal demand highlights the need for:

- Effective demand forecasting, and
- Pre-season inventory planning to avoid lost revenue opportunities or excess stock.

Sales

This section looks at sales performance from different angles to understand what drives demand, how customers behave, and where operations focus within the fast-fashion supply chain. Sales are measured by units sold and revenue, compiled from shipment-level log data.

Gender

Sales are mainly focused on men's apparel. Men's products make up about 74% of total units sold and about 77% of total revenue, making this the largest demand segment. Women's products bring in around 23% of revenue, while unisex items represent a very small portion.

This imbalance shows a male-centered product strategy, which increases risk when there are high-return or high-delay issues with men's SKUs, as later discussed in the returns analysis. Since demand is concentrated in men's categories, any issues with quality, fit, or forecasting in men's apparel have a significant financial impact on the overall supply chain.

Weight

Sales volume is mainly in medium-weight apparel, such as jackets, blazers, hoodies, and jeans. Lightweight accessories sell often, but they add less to revenue. Heavy products, like outerwear, sell fewer units but generate higher revenue per unit due to their higher prices. Weight indirectly affects logistics costs, but sales patterns depend more on seasonality and fashion trends than on weight alone. While weight may not be a primary factor for demand, it becomes important later regarding transportation and holding costs.

SKUs

Sales are concentrated among a small group of SKUs. A limited selection of men's apparel items, such as blazers, jackets, jeans, hoodies, and chinos, make up a significant portion of total sales. Many SKUs sell steadily but in much lower volumes, creating a long tail. This concentration in sales supports the later application of ABC analysis, which focuses management on a few high-impact products.

Ratings

Sales have a positive but not perfect connection with average batch ratings. Products that consistently get low ratings rarely reach high sales. However, some top-selling SKUs still have just moderate ratings. This indicates that brand strength and fashion relevance can be more important than quality perceptions. The impact of ratings is more evident in return behavior than in initial sales volume. Customer ratings influence the long-term sustainability of sales. Still, short-term demand in fast fashion can stay high, even with signs of moderate quality.

Price

Analysis of selling price in relation to units sold, revenue, and profit shows a clear trade-off between volume and margin. SKUs priced between 30 and 50 account for the largest share of total units sold. They represent the main demand group of the assortment. These products are usually everyday clothing that people buy repeatedly. Higher-priced SKUs, like jackets and blazers in the 70 to 80 range, sell fewer units. However, they generate much higher profits due to better per-unit contributions and favorable cost structures. Very low-priced items, priced at 15 or less, achieve high volume but contribute only a small amount to total profit. This highlights the need for a selective product range rather than focusing solely on maximizing volume.

Factory and Warehouse

Sales are unevenly spread across factories and warehouses. A few factories, especially F001 and F005, account for a large portion of total sales. Likewise, several warehouses regularly manage higher outbound volumes. This concentration in operations leads to scale efficiencies, but it also raises the risk of disruptions at these key points.

Returns

The returns analysis looks at where and why products are returned, using shipment-level data from the log dataset. Returns are measured as:

Return Rate = No. of Pieces Returned / No. of Pieces Sold

This method lets us compare products, factories, warehouses, and customer segments without assuming equal volumes.

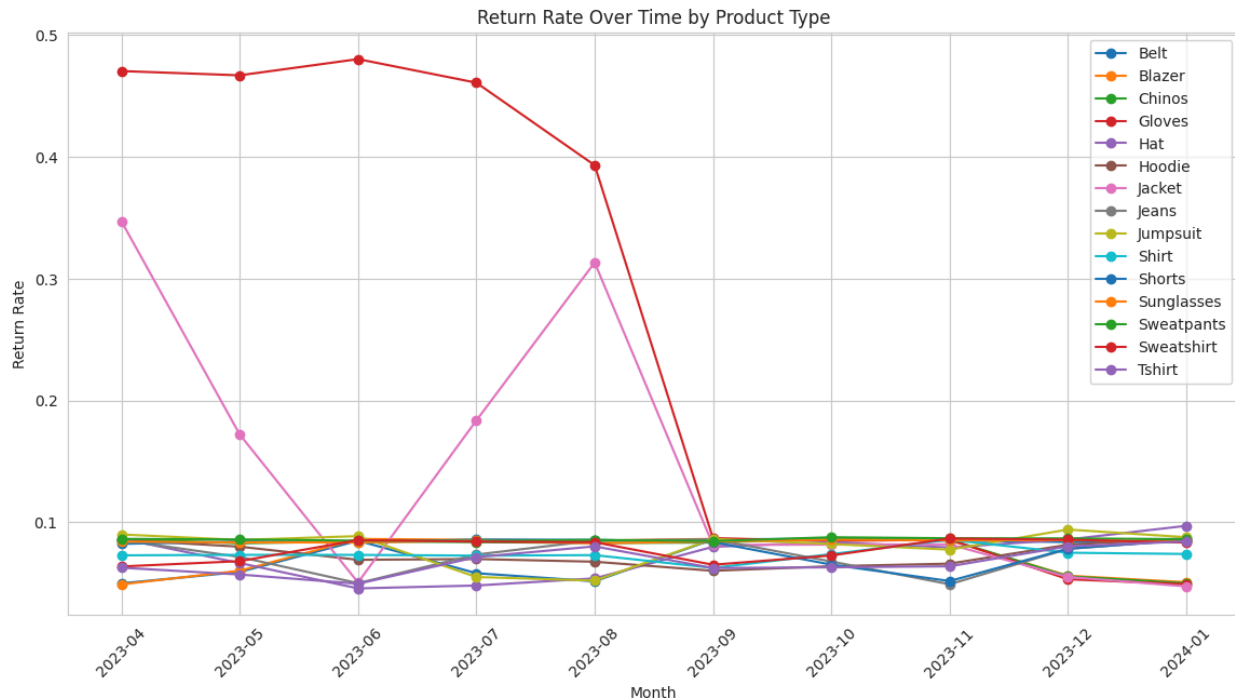
Gender

Although men's products generate most of the revenue, they also account for most of the returns. This suggests that leading in revenue does not mean there is operational efficiency.

Weight

There is no direct connection between weight and returns. Weight does not represent logistics damage or shipping problems. Returns are driven by consumers, not by how items are handled.

SKUs



The analysis shows clear return outliers. Men's Gloves have the highest return rate at about 28.7%, making them the most problematic item in the portfolio. Men's Jackets come next with a return rate of around 20.8%, which is much higher than the system average. In contrast, several women's and unisex products have return rates between 5 and 8%. This shows stable satisfaction after purchase. This sharp contrast suggests that the issue is not widespread across all categories, like outerwear. Instead, it focuses on specific men's items, indicating problems with fit, sizing, or quality consistency.

Ratings

There is no straightforward link between weight and returns. This means that the returned products are not always of lower quality; they may simply not meet the customers' expectations.

Price

Price shows a u-shaped return pattern, with slightly higher returns for both lower tier and higher tier priced items, most likely due to the pulls on either end from the gloves, and jackets. Otherwise, price does not have a significant impact on returns.

Factory and Warehouse

Return concentration is linked to specific factories instead of being spread out evenly. Factories F002 and F005 have more high-return SKUs, particularly Men's Gloves and Jackets. On the other hand, factories F001 and F003 handle large volumes but show lower return rates. This indicates they might have better process control or consistent quality. It highlights the need to

view returns as a supply-side issue, not just one related to demand. Meanwhile, warehouse-level return rates are quite steady, and no warehouse emerges as a major return location. This suggests that:

- Returns are not affected by last-mile delivery or regional handling.
- Root causes are earlier in the process, such as design, production, and quality control, rather than in logistics

Delays

Delays were computed using:

$$\text{Delay} = \text{Shipping Time (Actual)} - \text{Shipping Time (Expected)}$$

A shipment was marked delayed if

$$\text{Delay} > 0$$

Delay analysis was done across SKU, factory, warehouse, gender, weight, and distance. The system shows an overall delay rate of about 35 to 36%. This means that more than one in three shipments arrives later than expected.

However, the delays are usually small, typically within a narrow range of plus or minus 2 days. This indicates a pattern of frequent, but low-impact, delays.

Weight

Heavier items (e.g., Jackets, Blazers) do not experience higher delays and lightweight items do not show systematically faster delivery.

SKUs

Delay rates are similar across most SKUs. High-demand SKUs do not consistently show higher delays. Delays are thus system-wide operational issues.

Factory and Warehouse

Factory-level delay rates are very similar. No factory stands out as the main problem:

- The factory with the highest delay rate has rates only slightly above the system average.
- The factory with the lowest delay rate still faces significant delays.

This shows that:

- Delays are a system-wide issue, not specific to any one factory.
- Problems likely stem from planning, batching, or coordination, rather than from inefficiencies in a single factory.

Warehouse delay rates reflect factory patterns where no warehouse shows significant variation.

Distance

Distance analysis shows almost no connection between distance and delay occurrence.

Long-haul routes are not typically more delayed than shorter ones.

This confirms that:

- Delays are driven by processes, not by geography.
- Redesigning the network alone will not fix delay issues.
- Warehouse operational failures.

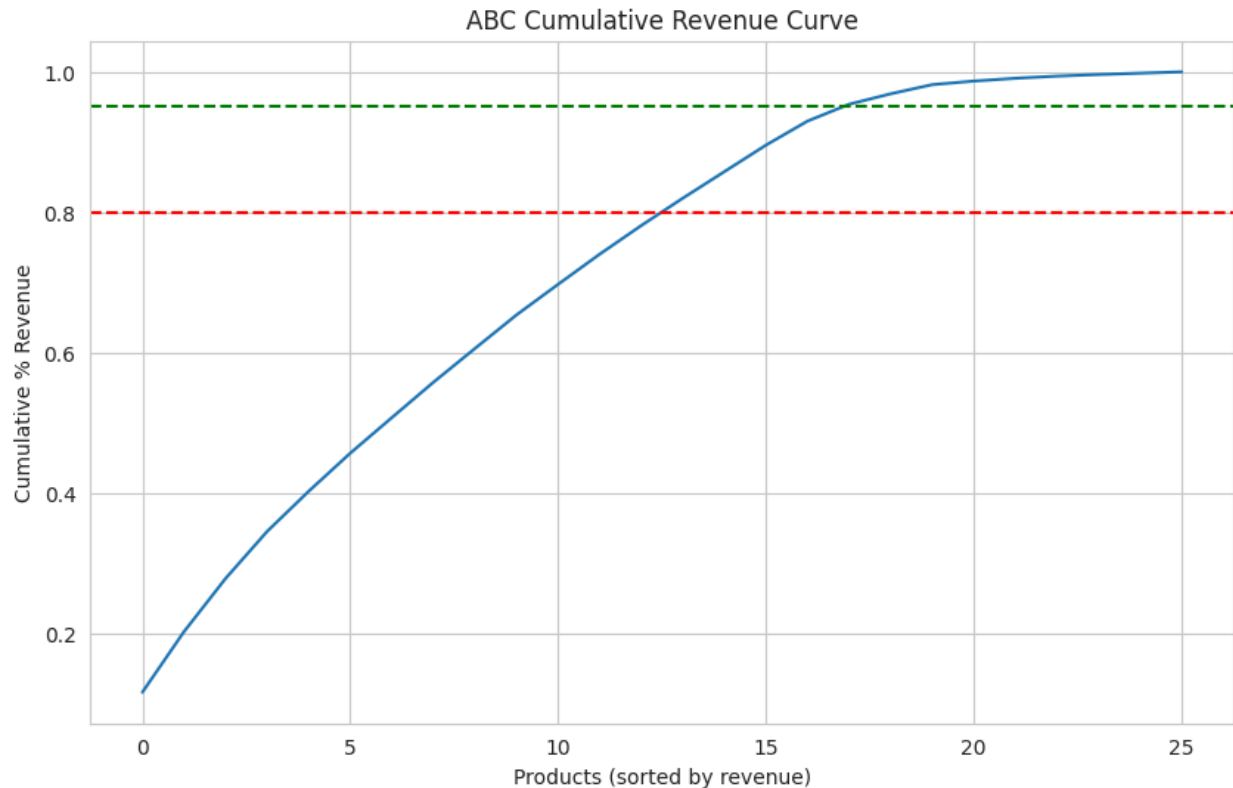
Methodology

ABC Analysis

Purpose of ABC Analysis

ABC analysis identified the products that significantly contribute to total revenue. This helps prioritize forecasting efforts, inventory planning, and management focus. With 40 SKUs and only 10 months of historical data, it was not practical or necessary to use complex forecasting models for all products.

Annual revenue for each SKU was calculated as the number of units sold into the selling price. The SKUs were then ranked in order of revenue, and cumulative revenue percentages were computed to categorize products into A, B, and C groups.



ABC Classification Results

A-Class products:

- 13 SKUs.
- These make up about 78 to 80% of total revenue.

B-Class products:

- 4 SKUs.
- These bring cumulative revenue to about 90 to 95%.

C-Class products:

- 9 SKUs.
- These contribute the remaining roughly 5% of revenue.

This shows a clear Pareto pattern: a small number of SKUs drives most of the financial performance.

The A-Class group mainly consists of men's apparel, especially:

- Structured outerwear (blazers, jackets).
- Bottoms (jeans, chinos).
- High-volume casual wear (hoodies, sweatshirts).

Unlike traditional cases where 20% of SKUs generate 80% of revenue, this dataset shows that about one-third of SKUs are needed to reach the 80% threshold. This indicates a broad product portfolio where various SKUs contribute significantly. Men's Blazers, Jackets, Jeans, and Chinos appear early in the cumulative curve. This signals strong demand coupled with higher price points. Most women's SKUs fall into B or C classes. This is not due to poor sales but because their absolute revenue contribution is lower compared to men's structured apparel.

Selection of SKUs for Demand Forecasting

While ABC analysis was the main filter, it was not the only factor used to choose products for detailed demand forecasting. The final selection considered three things:

- Revenue impact (ABC position)
- Demand behavior and seasonality
- Operational relevance (returns, volatility, category coverage)

The six SKUs chosen for forecasting were:

- Men's Gloves
- Men's Jacket
- Men's Blazer
- Women's Sweatshirt
- Men's Sunglasses
- Women's Sweatpants

Men's Jacket and Men's Blazer are firmly within A-Class, ranking among the highest contributors to revenue and profit. Forecasting these products is fully justified from a revenue-risk perspective.

Men's gloves and men's sunglasses are nearer to the B/C boundary in terms of revenue. However, both show high unit volumes for their price, which makes them important to operations even though they contribute less to revenue. These products also have significant return rates, so forecasting is useful for planning inventory and quality, not just for maximizing revenue.

The selected SKUs deliberately span:

- Outerwear (Jacket, Blazer)
- Casual wear (Women's Sweatshirt)
- Accessories (Gloves, Sunglasses)
- Bottoms / comfort wear (Women's Sweatpants)

Women's Sweatshirt and Women's Sweatpants do not fall within the top A-Class SKUs. Their inclusion was driven by:

- Stable demand patterns,
- Lower seasonal fluctuations,
- The need to prevent male-category bias in operational planning.

ABC analysis shows where financial risk is concentrated. However, forecasting decisions must also consider:

- Demand variability
- Returns behavior
- Category representation
- Operational complexity

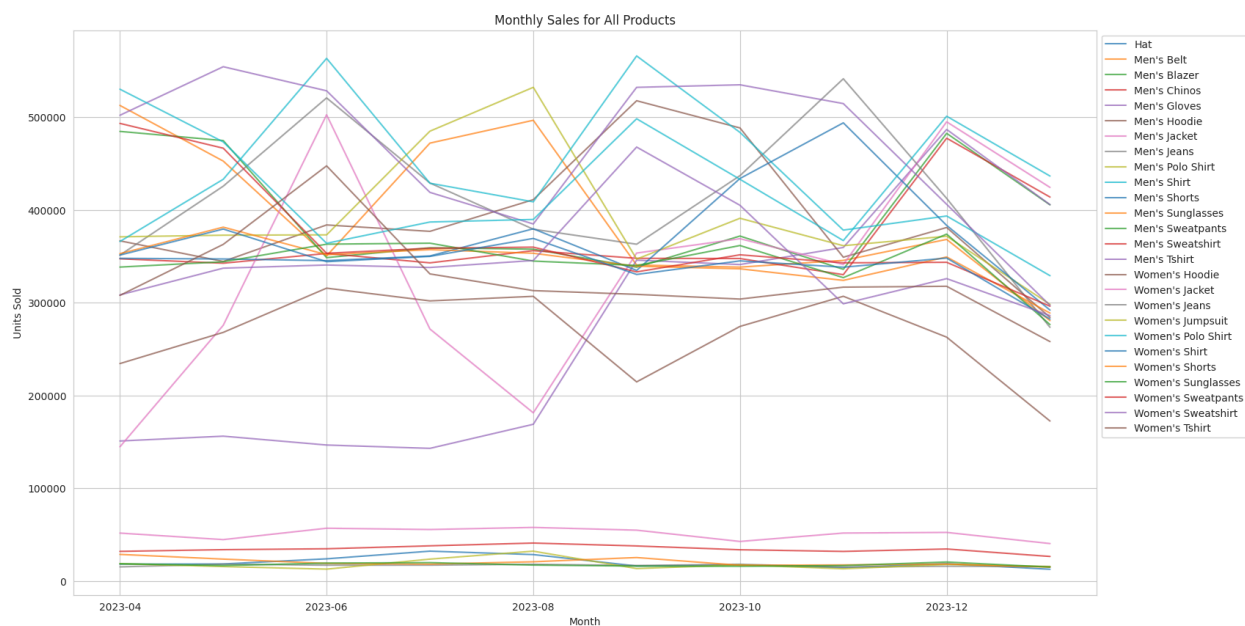
As a result, in this project forecasting focused on:

- A-Class SKUs for revenue protection
- Select B/C SKUs for operational control and gaining insights

Demand Forecasting

Objective

The aim is to find the most accurate forecasting method for each product by looking at actual error metrics.



Data Used

- Monthly demand aggregated from Log Data (No. of pieces sold)
- Time horizon: April 2023 – January 2024 (10 observations per SKU)
- For each SKU, the following models were evaluated:
 - Moving Average (MA-3, MA-6)
 - Weighted Moving Average (WMA-3, WMA-6)
 - Simple Exponential Smoothing (SES)
 - Holt's Linear Trend
 - ARIMA (baseline, non-seasonal)
- Forecasting evaluated on:
 - MAE
 - RMSE
 - MAPE (primary selection metric)

SKU	Product	Why Selected	Best Model	MAPE
P014	Men's Gloves	Strong winter seasonality + highest returns (28.7%)	Holt	8.84%
P010	Men's Jacket	Dual-season + volatile; high returns (20.8%)	Holt	15.5%
P009	Men's Sunglasses	Strong summer peak + declining trend	Holt	5.97%
P016	Women's Shirt	Stable baseline SKU; clothing essential	SES	14.0%
P022	Women's Sweatpants	Consistent winter lift; stable month-to-month	SES	11.7%
P028	Men's Jeans	Core apparel; A-class volume; needed a core non-seasonal men's SKU	MA-3	25.27%

Key Forecasting Insights

Forecast accuracy varied significantly across SKUs, showing that no single model works for all products. I used Mean Absolute Percentage Error (MAPE) as the main evaluation metric because it lets us compare products with different demand volumes. Among the six selected SKUs, the best MAPE values ranged from 5.97% to 25.27%. Trend-driven and seasonal products such as

Men's Gloves (8.84%), Men's Jackets (15.5%), and Men's Sunglasses (5.97%) had lower errors when we used Holt's Linear Trend model, indicating that these SKUs have predictable trends over time. In contrast, Men's Jeans showed the highest error at 25.27% with MA-3. This reflects its stable but noisy demand pattern, which lacks a strong trend or seasonality, making accurate forecasting challenging.

Smoothing-based models like Simple Exponential Smoothing (SES) performed better for stable or low-trend products, such as Women's Sweatpants (11.7%) and Women's Shirts (14.0%), where demand varied within a narrow range. ARIMA models consistently underperformed, often yielding MAPE values close to 100%. This happened because I only had a short data history of 10 months, which isn't enough for reliable parameter estimation. I also avoided classical seasonal decomposition methods since I didn't have a complete 12-month seasonal cycle, making seasonal indices unreliable. Instead, I used trend-aware models like Holt to implicitly capture seasonal effects when present.

Managerial Implications

- Forecasting accuracy improves significantly when model choice reflects demand behaviour
- High-impact, seasonal SKUs should use trend-sensitive models
- Stable products can rely on simpler smoothing techniques
- Avoiding over-complex models (like ARIMA) is often optimal when data history is short

The validated demand forecasts for these six SKUs were used to make decisions about inventory planning. This included setting safety stock, reorder points, and EOQ calculations. This method ensured that demand estimates matched the operational needs.

Inventory Planning

Objective

The goal of inventory planning was to find the best order quantities and restocking policies. This would help reduce total inventory costs while keeping service levels high for important SKUs.

Inventory planning was carried out using:

- Forecasted demand
- EOQ model
- Safety stock and reorder point calculations

Assumptions Used

The ordering cost is ₹60,000 per order. This value represents the total fixed cost of placing one replenishment order, including:

- Purchase order processing and administration
- Quality inspection and handling
- Inbound logistics and consolidation
- Coordination overhead during peak fashion seasons

For fast-fashion supply chains in India, ordering costs are high because of frequent restocking, tight deadlines, and inspection needs. A conservative but realistic estimate for medium-to-large batch orders is ₹60,000, especially since the business is still growing.

The ordering cost is 30% of unit cost. Holding cost includes:

- Cost of capital tied up in inventory,
- Warehousing and handling,
- Insurance and shrinkage,
- Markdown and obsolescence risk, which is especially high in fast fashion.

Because fashion products have short lifecycles and a high risk of markdown at the end of the season, holding costs are much higher than in traditional retail. A 30% rate reflects industry practice and errs on the side of caution.

EOQ Model and Results

Warehouse	Annual Demand	Unit Cost (₹)	Holding Cost (₹)	EOQ (units)	Total Inventory Cost (₹)
W001	216,870	2,009	603	6,570	3.96 M
W002	221,555	1,209	363	8,560	3.11 M
W003	231,924	3,609	1,083	5,070	5.49 M
W004	216,293	2,809	843	5,549	4.68 M
W005	443,328	409	123	20,810	2.56 M

- Low unit cost leads to large EOQs. W005 has the largest EOQ despite high demand because its unit cost and holding cost are very low.

- High unit cost constrains EOQ. W003 manages expensive inventory, which results in high holding costs and smaller EOQs, increasing the total inventory cost.
- EOQ balances holding and ordering costs as expected (Waters, 2003; Heizer et al., 2017). Across warehouses, annual holding costs and ordering costs are approximately equal, which confirms correct EOQ behavior.
- Warehouse-specific EOQs outperform a single global policy. Applying EOQ at the warehouse level captures cost differences and prevents overstocking at expensive locations.

Reorder Point (ROP) and Safety Stock

Reorder Point was calculated as:

$$\text{ROP} = (\text{Average Demand} \times \text{Lead Time}) + \text{Safety Stock}$$

with

- Average demand derived from forecasts
- Lead time assumed constant across warehouses
- Safety stock based on forecast error and demand variability

Seasonal SKUs like gloves and jackets require higher safety stock before winter. In contrast, stable SKUs such as women's sweatpants need little to no safety stock. Planning inventory based on forecasts significantly reduces stockouts and extra inventory. For high-demand, low-cost SKUs, bulk ordering helps keep ordering costs down. Expensive SKUs need more frequent replenishment cycles and tighter inventory control. It's essential to use realistic assumptions for ordering and holding costs in fast fashion, where inventory risk mainly comes from obsolescence.

Warehouse Selection

P-Median

The goal of the p-median analysis was to find the best group of warehouses to minimize the total demand-weighted distance between warehouses and customers. Doing so will lower transportation costs and improve service efficiency for the chosen high-impact SKUs.

The p-median model works well for fast-fashion supply chains because:

- Demand is spread out across different areas.
- Transportation cost increases with distance and volume.
- The company needs to run a limited number of warehouses (p) because of fixed infrastructure costs.
- Service speed is important, but distance is a measure of cost, not performance risk.

The demand used was the estimated demand for the final six selected SKUs. I combined and distributed it among warehouses based on past shipment patterns. I calculated the distance using realistic latitude and longitude coordinates for warehouses and demand centers, which reflect India's logistics situation. I set ($p = 5$) to focus on an effective regional distribution strategy instead of trying to cover the entire country with all available warehouses. This method is similar to how fast-fashion retailers operate, as they depend on a few regional hubs. It helps control transportation costs, improves responsiveness, and manages complexity. The p-median model then identifies the best five warehouses to minimize total demand-weighted distance across the network.

The selected p-Median Warehouses ($p=5$) are:

- W005
- W002
- W019
- W010
- W013

with an objective function value of 2,222,584,128 representing the minimum achievable total demand-weighted distance under the constraint of using only 5 warehouses.

Centre of Gravity

The Center of Gravity analysis aimed to find one ideal warehouse location that minimizes the total demand-weighted distance in the supply chain network. Unlike the p-median model, which selects several specific warehouse sites, the CoG provides a continuous location benchmark that shows the average demand pull across the country.

To calculate the Center of Gravity, I used demand-weighted latitude and longitude coordinates. The demand values for this calculation came from the predicted demand for the final six selected SKUs. I combined these values and spread them across warehouses based on historical shipping proportions. I created geographic coordinates with a synthetic method, but they were based on actual logistics hubs in India, ensuring realistic spatial relationships.

The calculated Center of Gravity was found in the Nagpur region, roughly at:

- Latitude ≈ 21.14
- Longitude ≈ 78.20

Nagpur is recognized as a major logistics center in India. It lies at the point where important north-south and east-west freight corridors meet. The CoG result aligns with real-world logistics planning. In this context, centrally located hubs help gather inventory and reduce transportation distances

Importantly, this result does not mean the company should rely on just one warehouse. Instead, the CoG acts as:

- A standard for network centrality
- A confirmation of the p-median solution
- A strategic reference for long-term network redesign

The CoG shows that the demand for the chosen SKUs is spread out evenly across different regions, rather than being heavily concentrated in one area.

Multi-Criteria Decision Making

While the p-median model finds the best set of warehouses by focusing only on minimizing demand-weighted distance, it overlooks other important factors like cost efficiency, delay risk, or return behavior. To support the p-median approach, I did a multi-criteria decision analysis (MCDA) to rank each warehouse based on various performance factors that matter for fast-fashion supply chains.

A TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) approach was used to rank warehouses.

Each warehouse was assessed based on these criteria:

Criterion	Description	Direction
Total Demand	Aggregate demand served by the warehouse	Higher is better
Distance from Center of Gravity	Distance to the network's CoG	Lower is better
Average Unit Cost	Avg. production + shipping cost	Lower is better
Delay Rate	Fraction of delayed shipments	Lower is better
Return Rate	Fraction of returned units	Lower is better

All criteria were normalized using min-max scaling. Cost, distance, delay rate, and return rate were considered as cost criteria and inverted during normalization. Demand was treated as a benefit criterion.

The final TOPSIS score shows how close each warehouse is to the ideal solution, which includes high demand, low distance, low cost, and low risk.

Warehouse	Demand	Dist. to CoG (km)	Unit Cost	Delay Rate	Return Rate	TOPSIS Score	Rank
W017	567,724	298.4	3,054.6	0.270	0.080	0.809	1
W012	556,880	539.1	3,054.6	0.269	0.081	0.717	2
W011	571,627	616.1	2,194.6	0.271	0.077	0.714	3
W001	565,253	575.4	3,694.6	0.275	0.081	0.686	4
W015	559,483	693.4	3,054.6	0.263	0.079	0.659	5
W005	672,780	822.5	2,813.5	0.253	0.081	0.644	6
W006	553,769	826.5	2,194.6	0.259	0.080	0.629	7
W020	679,758	864.3	3,366.8	0.265	0.081	0.611	8
W009	566,057	418.5	6,434.6	0.276	0.081	0.611	9
W013	552,518	92.6	8,054.6	0.267	0.080	0.606	10

W017, W012, and W011 are the top-ranked warehouses.

Why this matters:

- They handle moderate-to-high demand.
- They are at a reasonable distance from the CoG.
- They keep costs under control.
- They have average delay and return rates, with no outliers.

These warehouses may not be the cheapest or the closest, but they perform well across all areas.

Warehouses like W005 and W020 handle the highest demand, but their TOPSIS rank drops because they are farther from the CoG. Their combined cost and risk profile is also weaker than that of the top-ranked warehouses. This shows that demand alone is not enough for choosing the right warehouse.

W013 is extremely close to the CoG (92 km) but ranks last among the top 10 because:

- It has the highest unit cost
- Moderate delay and return rates

This shows that being near does not offset high operating costs when considering multiple factors.

Although delay and return rates fall within a narrow range (about 25–28% delays, 7.7–8.1% returns), TOPSIS still captures their combined effect.

In fast fashion, even small increases in delays or returns can significantly affect operations at high volumes. Considering these factors makes warehouse decisions more dependable.

p-Median finds the best group of warehouses to reduce demand-weighted distance. TOPSIS MCDA ranks warehouses based on cost, risk, and demand factors. Warehouses like W017, W011, and W012 do well in both methods. This highlights their strategic importance.

The MCDA and p-median analyses reveal where the network should function. The next step is to decide how to distribute product flows among factories and warehouses. This aims to lower total cost while staying within capacity limits.

Transport Optimization

The transportation optimization module was designed to identify the most cost-effective method for directing product flows from factories to warehouses while accounting for realistic operational limits. A linear programming (LP) formulation helped minimize total logistics costs throughout the network. The objective function aimed to reduce the total of manufacturing costs and transportation costs for all combinations of factories, warehouses, and SKUs, weighted by the quantity shipped. This method mirrors the operations of fast-fashion supply chains. Small cost differences among factories and routes can greatly affect profitability on a large scale.

Inputs to the Optimization Model

- Demand was based on the forecasted volumes of the final six selected SKUs. Forecasted demand was split among warehouses based on historical shipment patterns. This maintained consistency with how operations usually work.
- Two cost components were included:
 - Production cost: This includes specific manufacturing costs for each factory and SKU from the production cost dataset.
 - Transportation cost: This covers shipping costs from factory to warehouse for each SKU. The costs were adjusted to reflect realistic distance-based logistics using synthetic geographic coordinates.

- Factory Capacity Constraints: The dataset did not show clear limits on factory capacity. To avoid unrealistic solutions, I estimated capacities by using each factory's historical production volume and adding a 20% buffer.
- Non-negativity and feasibility constraints: All shipment quantities had to be non-negative, and total shipments from each factory could not exceed its estimated capacity.

The LP objective minimized total cost subject to the forecasted demand for each warehouse–SKU combination being met, factory capacity constraints, and non-negativity of shipment quantities.

The optimized solution resulted in a total logistics cost of about ₹227.8 million for the six-SKU planning period. This cost is not expected to match the historical logistics costs found in the raw operational data.

There are three main reasons for this difference:

- First, the historical data does not clearly reflect spatial distance. Transportation costs in the original dataset appear as transactional values, without considering geographic factors. In contrast, the LP model uses synthetic distances between factories and warehouses. This approach adds cost penalties for long-haul routes that were not easily seen before.
- Second, the optimization includes capacity limits, which restrict the volume processed through the cheapest factory. Historically, production might have been unlimited or based on short-term decisions instead of overall efficiency. Capacity limits force the model to take less economical routes once the cheaper options are used up, which increases the total cost.
- Third, the optimized cost reflects a structurally realistic future network, not past operating behaviour.

For these reasons, the optimized cost should be viewed as a strategic planning standard, not a direct comparison to historical revenue or profit.

The optimized solution showed clear structural patterns. Factory F001 was consistently selected as the main production source for high-volume SKUs. This indicates it provides the best mix of manufacturing cost and transportation efficiency. Factories F003 and F004 were used selectively, mainly when F001 was at full capacity or when there were specific cost benefits.

The optimizer completely avoided high-cost factory-warehouse-SKU routes. This is an important finding. These routes highlight structural inefficiencies that should be deprioritized or removed from regular planning unless they are needed for emergencies.

The model also favored fewer, larger shipments instead of many small ones. This approach aligns with EOQ-based inventory choices and reduces per-unit logistics costs.

Results and Conclusions

This project shows how supply chain analytics can be used not just to describe performance but to improve decision-making in merchandising, sourcing, inventory, and logistics in fast fashion. Instead of treating forecasting, inventory planning, and network design as separate tasks, the analysis combines them into a decision framework that acknowledges short product lifecycles, high return risks, and varied demand.

Descriptive Analytics

The exploratory analysis revealed that value creation is highly concentrated. A small number of SKUs brings in most of the revenue and profit. Many products add complexity to the assortment but offer little financial benefit. This discovery should change the way management approaches the situation. Rather than planning the same way for all products, the supply chain should focus on high-impact SKUs.

Actionable decision:

Management should establish SKU-tiered planning. A-class products should get dedicated forecasting models, higher service levels, and priority in warehouse and transport planning. Meanwhile, C-class products should use simpler planning methods, lower safety stock, and rely, when possible, on make-to-order or limited-batch strategies.

Demand Forecasting

The forecasting results show an important insight in fast fashion: no one forecasting model works for all products. For example, items with strong winter seasonality, like Men's Gloves and Men's Jackets, needed models that account for trends. In contrast, stable categories such as Women's Sweatpants benefited from smoothing-based methods. Models like ARIMA did not offer any advantages. This was mainly because of the short data history and the unpredictable nature of fashion demand.

Actionable decision:

Forecasting should be seen as a product-specific skill rather than a one-size-fits-all approach. Merchandising and supply planning teams should classify SKUs by demand behavior, trend-driven, seasonal, stable, or volatile, and assign models accordingly. This

also means different review cycles including regular re-forecasts for volatile winter SKUs and monthly reviews for stable items.

Inventory Planning

Inventory optimization presented a tough but genuine trade-off. Keeping inventory in fast fashion is expensive due to markdowns and the risk of items becoming outdated.

EOQ results indicated that warehouses managing low-cost products can and should make larger, less frequent orders. In contrast, those handling expensive inventory need to use smaller, more frequent replenishment cycles.

Actionable decision:

Inventory policy should be customized for each warehouse instead of being applied universally. Central planning should enable different EOQs and reorder points based on the warehouse and SKU. In practice, this means letting regional supply planners place bulk orders for low-cost basics. They will also closely oversee replenishments for high-value or high-risk items.

Network Design

The p-median and Center of Gravity analyses made the network's structure clear. India's geography supports a central-plus-regional hub model. Locations like Nagpur are emerging as strong consolidation points, backed by several high-capacity regional warehouses.

The MCDA (TOPSIS-based) analysis added an important perspective. Some warehouses may look good solely on demand or cost but can be less favorable once delay risks and return rates are considered. This reinforces that logistics decisions cannot rely on cost alone.

Actionable decision:

Management should separate strategic network design from operational routing. Strategic choices, like the number of warehouses and hub locations, should be reviewed each year using demand-weighted distance and risk metrics. Operational transport decisions should be improved continuously within this fixed network. Trying to optimize everything at once complicates matters and does not improve results.

Transportation Optimization

Transportation optimization led to costs that are higher than previous figures. This result is both expected and important. The optimized costs include distance, capacity limits, and practical routing, which were not clearly considered in earlier operational data. The updated cost reflects a future-oriented perspective on the supply chain..

Actionable decision:

Transportation optimization results should not be compared to past P&L items. They should instead answer strategic questions:

- Which factory-warehouse links should be expanded or eliminated?
- Where does capacity investment decrease long-term cost risks?
- Which routes should be avoided in future planning?

This views transportation optimization as a choice about network design instead of only a way to cut costs in the short term.

Strategic vs Tactical: Clarifying the Operating Model

A key outcome of this project is a clearer separation between strategic and tactical supply chain decisions:

- Strategic (6–18 month horizon):
 - Network structure, warehouse selection, factory allocation, SKU portfolio focus, service level targets(Chopra & Meindl, 2020).
- Tactical (weekly–monthly):
 - Forecast updates, replenishment quantities, shipment consolidation, short-term inventory adjustments.

Mixing these layers can lead to reactive decision-making and increased costs. Handling them separately allows the organization to stay flexible without the need to constantly redesign the system.

What the Company Should Do Next

- Incorporate SKU-tiering (ABC + demand behavior) into planning governance.
- Formalize differentiated forecasting ownership by product type.
- Implement warehouse-specific inventory policies rather than global rules.
- Establish a regional hub-and-spoke network and optimize within it.
- Use optimization results as strategic guidance

These steps shift the company from reacting to events to creating a strong fast-fashion supply chain. This new structure can handle changes while protecting speed, profit margins, and customer experience.

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